

1 Partielles Ableiten

Bilde jeweils die partielle Ableitung nach x und nach y.

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|--|--|
| a) $g(x, y) = \frac{xy}{4+2y^2}$ | e) $f(x, y) = \log(e^x) + e^{4y}$ |
| b) $f(x, y) = \log(xy)$ | f) $m(x, y) = x^3 \cdot \log(y)$ |
| c) $h(x, y) = a \cdot \sin(x) + \cos(y)$ | g) $f(x, y) = e^{x^2-y}$ |
| d) $p(x, y) = x - \log(y^2 + x)$ | h) $h(x, y) = \log(y) + y \cdot \log(x)$ |

2 Partielle Integration

Bestimme die Stammfunktion oder das genaue Integral (falls Grenzen angegeben sind) der jeweiligen Funktion.

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|--|--------------------------------|
| a) $\int x \cdot \ln(x) dx$ | c) $\int e^x \cdot \sin(x) dx$ |
| b) $\int_0^{\frac{\pi}{2}} x \cdot \cos(x) dx$ | |

*Tipp für c) Nutze zweimal partielle Integration. Nachdem du das zweite Mal partiell integriert hast, betrachte den Ausdruck vorsichtig.

3 Integration mit Substitution

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|-------------------------------|-------------------------------|
| a) $\int_0^3 3x^2 e^{x^3} dx$ | c) $\int \frac{\ln(x)}{x} dx$ |
| b) $\int x \sqrt{1+x^2} dx$ | |

* Tipp für c): Setze hier $u = 1 + x^2$

1 Partielles Ableiten

$$a) \quad \frac{\partial g(x,y)}{\partial x} = \frac{y}{4+2y^2}$$

$$\frac{\partial g(x,y)}{\partial y} = \frac{(4+2y^2)x - (xy \cdot 4y)}{(4+2y^2)^2} = \frac{4x + 2xy^2 - 4xy^2}{(4+2y^2)^2} = \frac{4x - 2xy^2}{(4+2y^2)^2} \quad \frac{NAZ - ZAN}{N^2}$$

$$b) \quad \frac{\partial f(x,y)}{\partial x} = \frac{1}{xy} \cdot y = \frac{1}{x} \quad \frac{\partial f(x,y)}{\partial y} = \frac{1}{xy} \cdot x = \frac{1}{y}$$

$$c) \quad \frac{\partial h(x,y)}{\partial x} = a \cdot \cos(x) \quad \frac{\partial h(x,y)}{\partial y} = -\sin(y)$$

$$d) \quad \frac{\partial p(x,y)}{\partial x} = 1 - \frac{1}{y^2+x} \quad \frac{\partial p(x,y)}{\partial y} = -\frac{1}{y^2+x} \cdot 2y$$

$$e) \quad \frac{\partial f(x,y)}{\partial x} = \frac{1}{e^x} \cdot e^x = 1 \quad \frac{\partial f(x,y)}{\partial y} = 4 \cdot e^{4y}$$

$$f) \quad \frac{\partial m(x,y)}{\partial x} = 3x^2 \cdot \log(y) \quad \frac{\partial m(x,y)}{\partial y} = \frac{x^3}{y}$$

$$g) \quad \frac{\partial f(x,y)}{\partial x} = e^{x^2-y} \cdot 2x \quad \frac{\partial f(x,y)}{\partial y} = -e^{x^2-y}$$

$$h) \quad \frac{\partial h(x,y)}{\partial x} = \frac{y}{x} \quad \frac{\partial h(x,y)}{\partial y} = \frac{1}{y} + \log(x)$$

$$a) \int x \cdot \ln(x) dx \quad \int u(x)v'(x)dx = u(x)v(x) - \int u'(x)v(x)dx$$

$$v'(x) := x \quad \Rightarrow \quad v(x) = \frac{1}{2} x^2$$

$$u(x) := \ln(x) \quad \Rightarrow \quad u'(x) = \frac{1}{x}$$

$$\begin{aligned} \int x \cdot \ln(x) dx &= \ln(x) \cdot \frac{1}{2} x^2 - \int \frac{1}{x} \cdot \frac{1}{2} x^2 dx \\ &= \ln(x) \cdot \frac{1}{2} x^2 - \int \frac{1}{2} x dx = \ln(x) \cdot \frac{1}{2} x^2 - \left[\frac{1}{4} x^2 \right] \end{aligned}$$

$$\Rightarrow \int x \cdot \ln(x) dx = \frac{x^2}{2} \ln(x) - \frac{x^2}{4} + C$$

Bestimmtes mit Grenzen $[1, 2]$:

$$\frac{x^2}{2} \ln(x) - \frac{x^2}{4} \Big|_1^2 = \left(\frac{4}{2} \ln(2) - 1 \right) - \left(\frac{1}{2} \ln(1) - \frac{1}{4} \right) = 2 \ln(2) - 1 + \frac{1}{4} = 2 \ln(2) - \frac{3}{4}$$

$$b) \int_0^{\frac{\pi}{2}} x \cdot \cos(x) dx \quad \int u(x)v'(x)dx = u(x)v(x) - \int u'(x)v(x)dx$$

$$v'(x) = \cos(x) \quad \Rightarrow \quad v(x) = \sin(x)$$

$$u(x) = x \quad \Rightarrow \quad u'(x) = 1$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} x \cdot \cos(x) dx &= x \cdot \sin(x) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 1 \cdot \sin(x) dx \\ &= \left(\frac{\pi}{2} \cdot \sin\left(\frac{\pi}{2}\right) - 0 \cdot \sin(0) \right) - \left(-\cos(x) \Big|_0^{\frac{\pi}{2}} \right) \\ &= \frac{\pi}{2} \cdot 1 - \left(-\cos\left(\frac{\pi}{2}\right) - (-\cos(0)) \right) = \frac{\pi}{2} - (0 - (-1)) = \frac{\pi}{2} - 1 \end{aligned}$$

$$c) \int e^x \cdot \sin(x) dx \quad \int u(x)v'(x)dx = u(x)v(x) - \int u'(x)v(x)dx$$

$$v'(x) = e^x \quad \Rightarrow \quad v(x) = e^x$$

$$u(x) = \sin(x) \quad \Rightarrow \quad u'(x) = \cos(x)$$

$$\int e^x \sin(x) dx = \sin(x) \cdot e^x - \int \cos(x) \cdot e^x dx$$

Betrachte:

$$\int \cos(x) \cdot e^x dx$$

$$v'(x) = e^x \quad \Rightarrow \quad v(x) = e^x$$

$$u(x) = \cos(x) \quad \Rightarrow \quad u'(x) = -\sin(x)$$

$$\int \cos(x) \cdot e^x dx = \cos(x) e^x - \int -\sin(x) \cdot e^x dx = \cos(x) e^x + \int \sin(x) e^x dx$$

Zusammen:

$$\int \sin(x) e^x dx = \sin(x) e^x - (\cos(x) e^x + \int \sin(x) e^x dx)$$

$$\int \sin(x) e^x dx = \sin(x) e^x - \cos(x) e^x - \int \sin(x) e^x dx$$

$$2 \cdot \int \sin(x) e^x dx = \sin(x) e^x - \cos(x) e^x$$

$$\int \sin(x) e^x dx = \frac{1}{2} (\sin(x) e^x - \cos(x) e^x) + C$$

3) Integration durch Substitution

a) $\int_0^3 3x^2 \cdot e^{x^3} dx$

1. Substitution

$$u = x^3 \quad \frac{du}{dx} = 3x^2 \quad \Rightarrow \quad du = 3x^2 dx$$

2. Einsetzen

$$\int_0^3 e^u \cdot du$$

3. Integration

$$\int_0^3 e^u du = e^u + C$$

4. Rücksub.

$$e^u + C = e^{x^3} + C$$

5. Grenzen einsetzen

$$e^{x^3} \Big|_0^3 = e^{3^3} - e^{0^3} = e^{27} - e^0 = e^{27} - 1$$

b) $\int \frac{\ln(x)}{x} dx$

1. Substitution

$$u = \ln(x) \quad \frac{du}{dx} = \frac{1}{x}$$

2. Einsetzen

$$\int u du$$

3. Integration

$$\frac{1}{2} u^2 + C$$

4. Rücksubstitution

$$\frac{1}{2} \ln(x)^2 + C$$

c) $\int x \cdot \sqrt{1+x^2} \, dx$

1. Substitution

$$u = 1+x^2 \quad \frac{du}{dx} = 2x \quad \Rightarrow \quad du = 2x \, dx \Leftrightarrow \frac{1}{2} du = x \, dx$$

2. Einsetzen

$$\int \sqrt{1+x^2} \cdot x \cdot dx = \int \sqrt{u} \cdot \frac{1}{2} du = \frac{1}{2} \int \sqrt{u} \, du$$

3. Integration nach u

$$\frac{1}{2} \int \sqrt{u} \, du = \frac{1}{2} \left(\frac{2}{3} u^{\frac{3}{2}} \right) + C = \frac{1}{3} u^{\frac{3}{2}} + C$$

4. Rücksubstitution $u = 1+x^2$

$$\frac{1}{3} u^{\frac{3}{2}} + C = \frac{1}{3} (1+x^2)^{\frac{3}{2}} + C$$